<u>TITLE</u>: HIGH-RESOLUTION CRYO-EM STRUCTURES OF VDAC2–BAX/BAK COMPLEXES TO ELUCIDATE MITOCHONDRIAL APOPTOSIS MECHANISMS.

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ABSTRACT

Apoptosis is a tightly regulated form of programmed cell death that plays a fundamental role in development and tissue homeostasis. In the intrinsic apoptotic pathway, cellular stress signals trigger the activation of the pro-apoptotic proteins BAX and BAK. These proteins translocate to the mitochondrial outer membrane, where they oligomerize to form pores that enable the release of cytochrome c, initiating the caspase cascade and ultimately leading to cell death. However, the molecular mechanism governing the recruitment and insertion of BAX and BAK into the mitochondrial outer membrane remains poorly understood. The Voltage-Dependent Anion Channel protein VDAC2, located in the mitochondrial outer membrane, has been proposed to play a key role in this process. Previous electrophysiological studies and co-purification experiments suggest a direct interaction between VDAC2 and BAX. In our study, VDAC2 is produced in nanodiscs, and the VDAC2–BAX complex is generated using a cell-free expression system. Following purification through the His-tag on VDAC2, immunoblot analysis confirms the presence of both VDAC2 and BAX. Cross linking experiments further support the formation of a stable complex between these proteins. Preliminary cryo-electron microscopy analyses are currently underway to visualize this complex at high resolution. Our objective is to characterize the structural basis of the VDAC2–BAX interaction to elucidate the molecular mechanism regulating mitochondrial membrane permeabilization during apoptosis. Understanding this interaction could provide new therapeutic opportunities to modulate apoptosis, either by inhibiting cell death in neurodegenerative diseases or by promoting

apoptosis in cancer cells. Structural insights into the complex may also enable future structure-based drug design targeting this pathway.

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